vantage points: the laws of physics appear identical to all observers undergoing constant-velocity motion. But this is limited democracy indeed, for it excludes an enormous number of other viewpoints—those of individuals who are accelerating. Einstein's 1907 insight now shows us how to embrace all points of view—constant velocity and accelerating—within one egalitarian framework. Since there is no difference between an accelerated vantage point without a gravitational field and a nonaccelerated vantage point with a gravitational field, we can invoke the latter perspective and declare that all observers, regardless of their state of motion, may proclaim that they are stationary and "the rest of the world is moving by them," so long as they include a suitable gravitational field in the description of their own surroundings. In this sense, through the inclusion of gravity, general relativity ensures that all possible observational vantage points are on equal footing. (As we shall see later, this means that distinctions between observers in Chapter 2 that relied on accelerated motion—as when George chased after Gracie by turning on his jet-pack and aged less than she—admit an equivalent description without acceleration, but with gravity.)

This deep connection between gravity and accelerated motion is certainly a remarkable realization, but why did it make Einstein so happy? The reason, simply put, is that gravity is mysterious. It is a grand force permeating the life of the cosmos, but it is elusive and ethereal. On the other hand, accelerated motion, although somewhat more complicated than constant-velocity motion, is concrete and tangible. By finding a fundamental link between the two, Einstein realized that he could use his understanding of motion as a powerful tool toward gaining a similar understanding of gravity. Putting this strategy into practice was no small task, even for the genius of Einstein, but ultimately this approach bore the fruit of general relativity. Achieving this end required that Einstein forge a second link in the chain uniting gravity and accelerated motion: the curvature of space and time, to which we now turn.

Acceleration and the Warping of Space and Time

Einstein worked on the problem of understanding gravity with extreme, almost obsessive, intensity About five years after his happy revelation in the Bern patent office, he wrote to the physicist Arnold